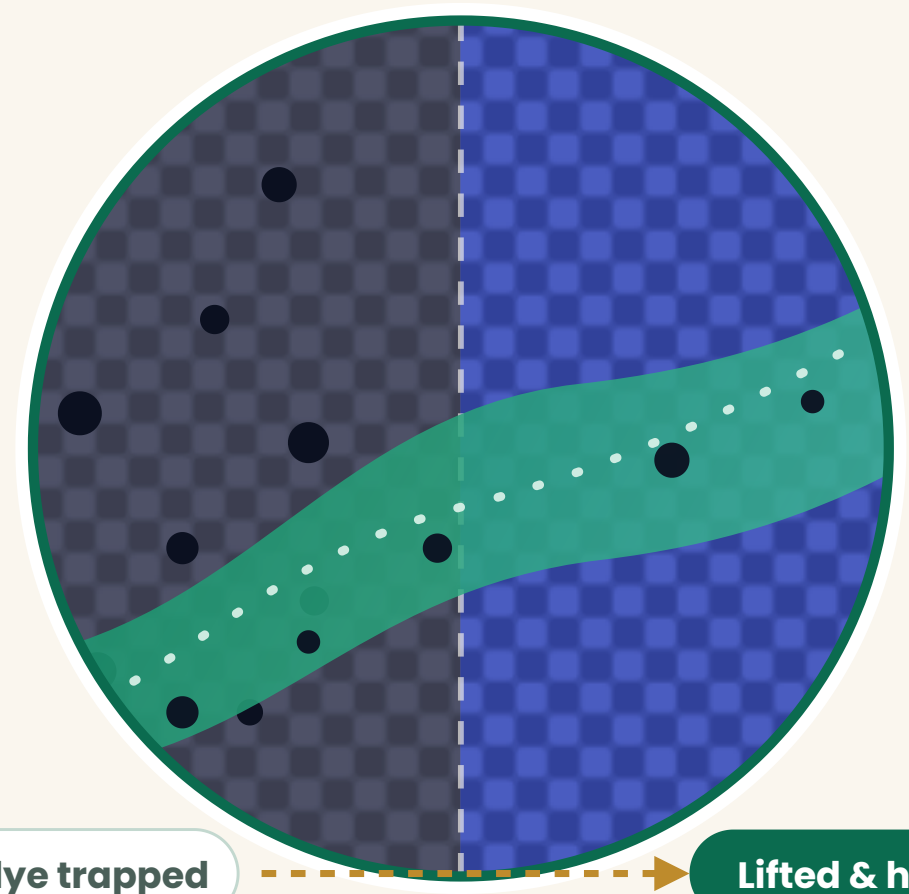
AEVA CHEMICALS

TEXTILE AUXILIARIES · REACTIVE DYEING



AeSop.

Low-temperature soaping agent for reactive-dyed cotton.

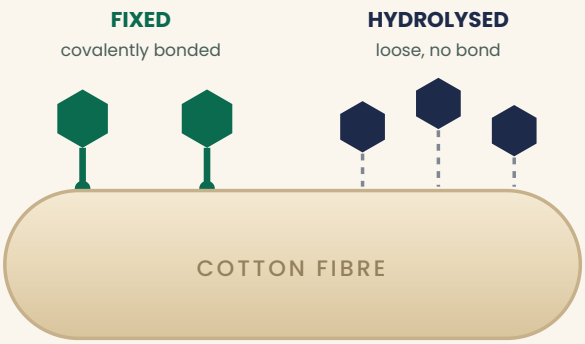
Polymer wash-off technology · Surfactant-free · Engineered for jet, soft-flow and winch machines

70°C	**3** baths	**4–5**	**0**%
Soaping temperature instead of 95°C	Fewer rinses to reach a clear bath	Multifibre staining grade on deep shades	APEO, NPEO and formaldehyde

Up to 40% of your reactive dye never bonds to the cotton.

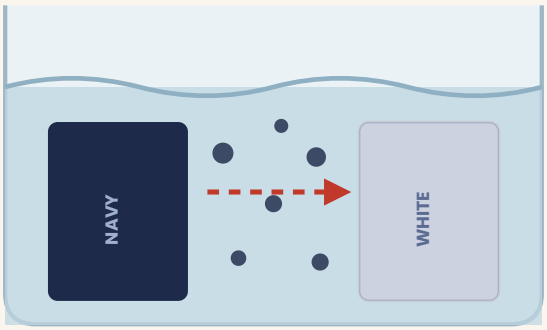
Reactive dyes hydrolyse in the alkaline dyebath. That hydrolysed dye cannot form a covalent bond — it simply clings to the fibre surface. Until it is removed, it will keep bleeding: onto whites, onto the customer's skin, onto the test strip in the buyer's lab.

A · At the fibre surface



Only the bonded dye is fastness.
The rest is a liability waiting to transfer.

B · Back in the rinse bath



Loose dye leaves the navy and lands on the white. This is backstaining.

Fastness failures

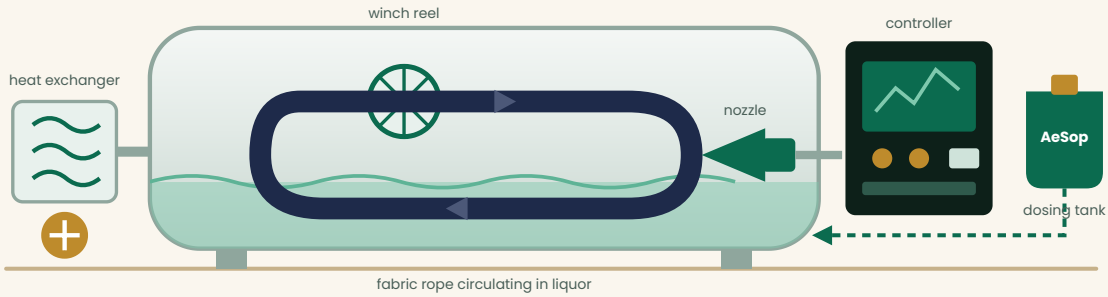
Wash, wet-rub and perspiration tests fail on navy, black and turquoise — the shades your best margins sit in.

Water and steam burnt

Six to eight rinse baths at 90–98°C, repeated on every machine, every shift — and effluent load to match.

Reprocessing

The most expensive outcome of all: a stripped and redyed lot, a late shipment, and a buyer who starts sampling elsewhere.



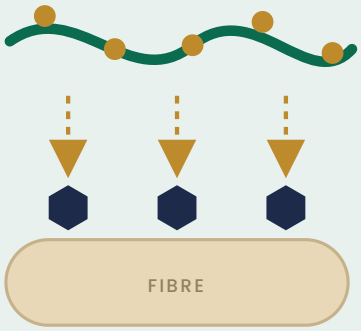
Soft-flow and jet machines are where wash-off is won or lost — high turbulence, short liquor, and salt carried in from the dyebath.

A dispersing polymer, not a detergent.

Conventional soapers are surfactant blends: they lift dye off the fibre but leave it loose in the bath, free to redeposit. **AeSop** is built on an anionic dispersing copolymer that captures hydrolysed dye and holds it in suspension until the bath is dropped.

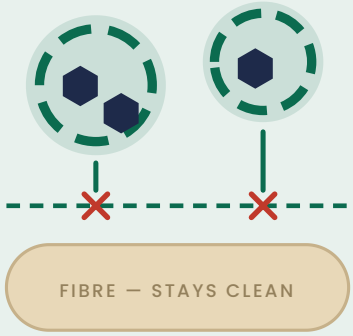
1 Capture

Anionic binding sites seek out hydrolysed dye



2 Disperse

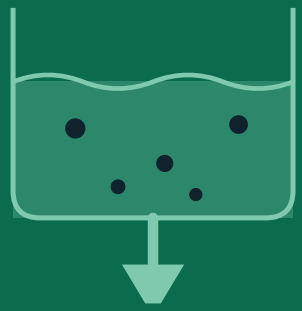
Dye is locked inside the coil and carries a negative charge



Redeposition blocked

3 Discharge

Dye leaves with the bath — not with the next rinse



Clear bath in fewer cycles

Engineered into the same molecule

Surfactant-free, low foam

No detergent base means no foam head under jet turbulence, and nothing left behind to interfere with the softener or finish applied next.

Sequestering built in

Chelating groups hold calcium and magnesium, so hard water does not blunt performance or throw insoluble dye-metal complexes onto the cloth.

Electrolyte tolerant

Keeps dispersing through the salt carried over from the dyebath, so the first wash-off bath works as hard as the last one.

Active from 60°C

Dispersion is not thermally driven, so the polymer performs at 60–75°C where a detergent soaper needs to be near boil.

One product, both ends of the market. AeSop is supplied as a single concentrated liquid. Run it at 1.0–1.5 g/L at 85–95°C to slot into an existing process and cut cost per batch, or at 0.8–1.2 g/L at 60–75°C to take the full water, steam and cycle-time saving. No second SKU, no re-training.

Rinse liquor, bath by bath

Wash-off of 6% o.w.f. navy on cotton knit, dosed with AeSop

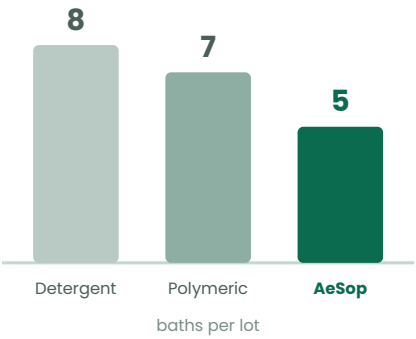


Bath five runs clear. A conventional soaper is still tinting the liquor at bath eight — which is what those extra baths are paying for.

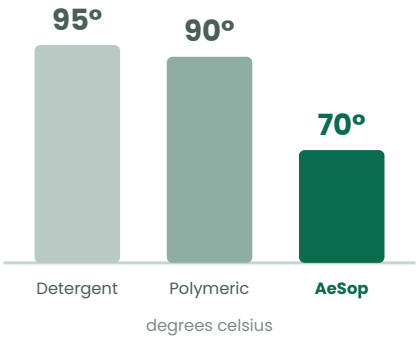
Against what is already on your floor.

Side-by-side wash-off on 6% o.w.f. navy reactive on knitted cotton, same machine, same dyeing recipe — AeSop against a conventional detergent-type soaper and a standard polymeric soaper.

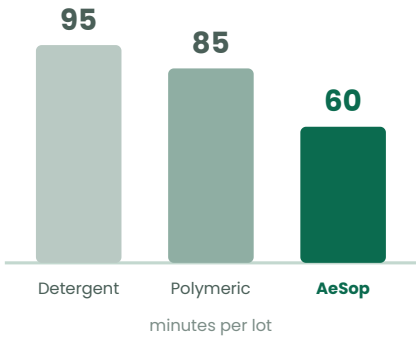
Rinse baths to a clear bath



Soaping temperature



Wash-off cycle time

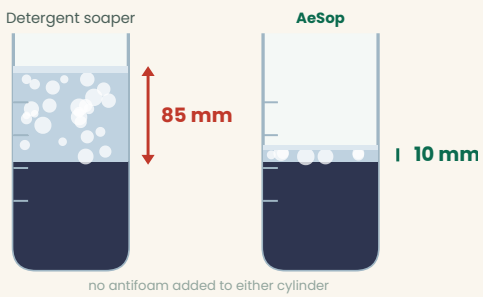


Where the three differ

Criterion	Detergent soaper	Standard polymeric	AeSop
Chemistry	Anionic/nonionic surfactant blend	Polymer + surfactant	Surfactant-free dispersing copolymer
Working temperature	90–98°C	85–95°C	60–95°C
Anti-backstaining	Limited	Good	Excellent — grade 4–5
Foam in jet machines	Moderate to high	Low to moderate	Practically nil
Hard water tolerance	Needs separate sequestrant	Partial	Sequestering built in
Salt carryover tolerance	Poor	Moderate	Up to 20 g/L
Residue / finish compatibility	Can affect absorbency	Generally neutral	No residue; softener-safe
Typical dosage	1.5–2.5 g/L	1.0–2.0 g/L	0.8–1.5 g/L

Foam height in the wash bath

1 g/L at 40°C, 30 seconds agitation



Why foam decides the sale in a jet machine

A foaming soaper in a high-turbulence machine costs you three times over. The rope rides on the foam and tangles or creases. The pump draws air and loses circulation, so the bath stops being uniform. And the antifoam you dose to fix it is itself a silicone that can spot the cloth and interfere with the softener afterwards.

AeSop carries no detergent base, so there is nothing to foam. No antifoam line item, no crease marks, no silicone spotting — and the pump runs the way it was specified to.

Indicative figures. Values shown are representative of laboratory and pilot-scale wash-off on reactive-dyed cotton and are given for comparison only. Actual results depend on shade depth, dye class, machine type, liquor ratio and water quality, and must be confirmed by trial at your plant.

Graded the way your buyer grades it.

Wash fastness is two separate judgements: has the shade itself changed, and has it stained anything next to it. A soaping agent is only doing its job when both come back clean.

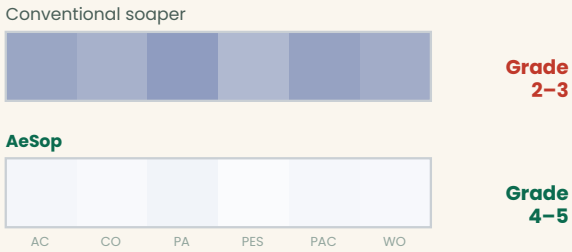
1 · Colour change

The dyed fabric itself, after wash · ISO 105-A02



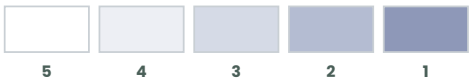
2 · Staining on adjacent fabric

Multifibre strip washed with the specimen · ISO 105-A03



Grey scale for staining

5 = no staining · 1 = heavy



Export buyers typically set a minimum of grade 4 on staining and 4 on colour change for cotton knitwear. Anything below that comes back.

Full fastness battery, 6% o.w.f. navy on cotton knit

TEST		CONVENTIONAL	AeSop
Wash fastness, colour change	ISO 105-C06	3–4	4–5
Wash fastness, staining on cotton	ISO 105-C06	2–3	4–5
Wash fastness, staining on polyamide	ISO 105-C06	2	4
Water, staining	ISO 105-E01	3	4–5
Perspiration, acid & alkaline	ISO 105-E04	3	4–5
Rubbing, dry	ISO 105-X12	3–4	4–5
Rubbing, wet	ISO 105-X12	2–3	3–4

One straight answer on wet rubbing: on extra-dark shades it is limited by dye chemistry, not by wash-off. Grade 3–4 is the realistic ceiling and no soaping agent will take you past it. Anyone promising 4–5 is selling you something.

3 · Backstaining on white ground

Yarn-dyed stripe, washed off in one bath with the dyed ground



Indicative figures. Grades shown are representative results on reactive-dyed cotton knit and are given for comparison. Fastness depends on dye selection, shade depth, fixation and finishing as well as wash-off, and must be confirmed on your own substrate and recipe. Test reports available on request.

Priced per litre. Judged per batch.

AeSop is not the cheapest soaping agent per kilogram, and it is not meant to be. It removes three baths, twenty-five degrees and thirty-five minutes from every dark-shade lot you run. That is where the money is.

Wash-off sequence, 6% o.w.f. navy on cotton knit

CONVENTIONAL · 8 baths

Drain	Cold rinse	Neutralise	Soap 1	Soap 2	Hot rinse	Hot rinse	Cold rinse
–	30°C	50°C	95°C	95°C	80°C	70°C	30°C

95 min

-37% time

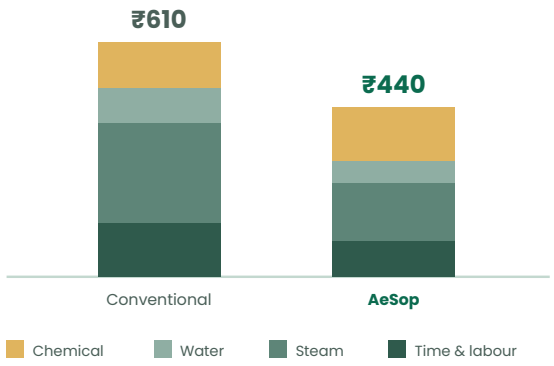
WITH AeSop · 5 baths

Drain	Cold rinse	Neutralise	AeSop soap	Warm rinse
–	30°C	50°C	70°C · 20 min	50°C

60 min

Cost per 100 kg of fabric

Indicative build-up, dark shade, batch dyeing



Chemical spend goes up. Everything under it comes down.

-32% — **Process water**
Three fewer fills and drops per lot

-40% — **Steam load**
70°C soap, one warm rinse instead of two hot

-37% — **Cycle time**
35 minutes back per machine, per lot

-28% — **Cost per batch**
Net of the higher chemical price

WHAT THAT ADDS UP TO

On a dyehouse running 3,000 kg a day

16,000 m³
of process water saved a year

40% less
steam drawn on the wash-off stage

1,750 hrs
of machine time released a year

Worked on 300 running days, an average lot of 300 kg, liquor ratio 1:6, and the sequence shown above. Substitute your own figures and the arithmetic holds.

Cost build-up is an illustrative model based on typical Indian batch-dyeing utility rates and is not a quotation. A costed comparison against your current soaping agent can be prepared using your own utility rates, machine data and shade mix.

AeSop — technical summary.

Appearance	Clear to pale amber liquid
Ionic character	Anionic
pH (1% solution)	6.5 – 8.0
Specific gravity (30°C)	1.05 – 1.15
Solubility	Miscible with water in all proportions
Working temperature	60 – 95°C
Liquor ratio	Effective from 1:4 upwards
Compatibility	Anionic and nonionic auxiliaries
Machinery	Jet, soft-flow, winch, jigger, continuous
Packaging	35 kg carboy · 120 kg drum · 1,000 kg IBC
Shelf life	12 months in sealed original packing
Storage	5 – 40°C, away from direct sunlight

Dosage by shade depth

Pale < 1% o.w.f.	**0.5 – 0.8 g/L**
Medium 1 – 3%	**0.8 – 1.0 g/L**
Dark 3 – 6%	**1.0 – 1.3 g/L**
Extra dark > 6%	**1.3 – 1.5 g/L**

Recommended procedure

After dyeing, drop the bath and give one cold rinse. Neutralise with acetic acid at 50°C. Fill fresh, dose AeSop, raise to 70°C and run 20 minutes. Drop, give one warm rinse at 50°C, and proceed to softening. No separate sequestrant or dispersant is required.

Compliance

GOTS approved input
Certificate available on request

ZDHC MRSL conformant
Gateway listing available on request

Full documentation pack
TDS, SDS, test reports, certificates

APEO & NPEO free
By formulation

Formaldehyde free
By formulation

Readily biodegradable
OECD 301 report on request

NEXT STEP

Evaluate AeSop against your current soaper.

Tell us your shade range, machine type, liquor ratio and water hardness and we will recommend a starting dosage and sequence. Sample quantity, technical data sheet, safety data sheet and certificates are issued together, and our application team supports the first production runs.



35 kg · 120 kg · 1000 k

AEVA CHEMICALS

Textile auxiliaries for reactive dyeing, manufactured in-house. Formulation, application support and technical service from one team.

AeSop™ · Certification status as shown at date of issue — confirm before quoting. Performance figures indicative and subject to plant trial.

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